

Shangyu Andrew Xu

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EDUCATION

Yale University

Aug 2023 – May 2027 (Expected)

B.S./M.S. in Computer Science, B.S. in Mathematics (Intensive)

GPA: 3.98/4.0

Coursework

ML (Grad-level): Natural Language Processing, Intermediate ML, Probabilistic ML, Deep Learning on Graphs

Math: Honors Linear Algebra, Honors Analysis, Abstract Algebra, Vector Analysis, Fields and Galois Theory

CS: Systems Programming, Spectral Graph Theory, Algorithms, Discrete Math

Activities & Awards: MathCounts Outreach, 5x AIME Qualifier, USAPhO Qualifier, USNCO Qualifier

EXPERIENCE

Machine Learning Researcher

Aug 2024 – Present

Natural Language Processing (NLP) Lab, Yale University

New Haven, CT

- Researching retrievers for reasoning-intensive tasks with Dr. Arman Cohan and Dr. Rex Ying
- Benchmarking top-performing retrievers on the Massive Text Embedding Benchmark (MTEB) on reasoning tasks
- Training retrievers in geometric space using PyTorch and evaluating on BEIR and BRIGHT benchmarks

Head Teaching Assistant

Aug 2024 – Present

CS223: Data Structures and Programming Techniques, Yale University

New Haven, CT

- Curating weekly section material on data structures and memory management in C for a 200-student class
- Developed 350+ public and private test cases for 6 brand-new coding projects in Fall 2024

Machine Learning Engineer Intern

May 2024 – Aug 2024

ActualFood (Startup)

Remote

- Built a full-stack iOS application in Swift focused on neural network-powered gesture recognition
- Designed and trained a CNN in PyTorch (inspired by MobileNetV4) for real-time hand gesture classification
- Quantized model to optimized on-device inference time for iPads

Business Manager

May 2024 – Present

Doox of Yale A Cappella

New Haven, CT

- Organized 70+ performances generating \$50k+ in revenue and 40 days of tour for 15 members across 3 continents
- Recorded 19 singers in a studio and edited, mixed, and mastered tracks with professionals for a 14 track [album](#)
- Leading the incorporation of Doox of Yale as a 501(c)(3) non-profit organization

PUBLICATIONS

Centralized Selection with Preferences in the Presence of Biases | *ICML 2024*

Nov 2023 – Jul 2024

[Paper](#) | [Slides](#) | [Code](#)

- Co-first author and presenter; authored with Dr. Nisheeth Vishnoi, Dr. Elisa Celis, and Dr. Amit Kumar
- Researched the mathematical optimization of a centralized selection algorithm under unknown bias factors
- Independently created 40+ simulations testing the robustness of core theorem under non-ideal conditions

PROJECTS

Graph Neural Networks for Fake News Detection | *PyTorch*

Dec 2024

[Project Proposal](#) | [Code](#)

- Capstone project for CPSC 483: Deep Learning for Graph-Structured Data
- Built GNN models with an attention-based architectures incorporating heterogeneous types and temporal data to detect fake news in graph-based datasets

TECHNICAL SKILLS

Languages: Python, C/C++, Swift

Machine Learning Tools: PyTorch, TensorFlow, Hugging Face Transformers, Microsoft Azure

Developer Tools: Git, VS Code, Xcode